FinsQuad

Ayana Sarkar, Kiran Rajar, Amy Chan

Abstract

We apply quantum reservoir computing (QRC) to the problem of implied volatility surface forecasting using the MerLin framework developed by Quandela. Swaptions — options on interest rate swaps — are widely used financial instruments whose pricing requires predicting the future evolution of a high-dimensional, structured object: the implied volatility surface. This makes the problem challenging for classical methods, as the surface is nonlinear, time-dependent, and often observed with missing or irregular data.

Our approach compresses the 224-dimensional volatility surface to a low-dimensional latent representation via principal component analysis, exploiting the fact that volume surfaces lie on a smooth low-rank manifold. These latent features are encoded into a photonic quantum circuit built with MerLin's CircuitBuilder, forming a frozen random reservoir whose weights are never trained. The reservoir maps inputs to a high-dimensional Fock state probability distribution through photon interference and entanglement, generating implicit polynomial feature expansions of the input. A classical linear readout is then trained on these quantum features to predict the next-day surface, keeping the training procedure convex and efficient.

We conduct a systematic ablation study isolating the contribution of each design choice: photon number, number of optical modes, entanglement depth, and reservoir ensembling. While classical Ridge regression remains competitive on this near-linear forecasting task, the ablation demonstrates that increasing Fock space dimensionality and entanglement strength consistently reduces prediction error, motivating further investigation into trainable encoding layers and larger photon numbers to improve the reservoir model for time series task like this and actually derive some advantage over classical methods in the future.

Main Notebook files are : **MerLin_Qvolution_swaptions_configs.ipynb**, **MerLin-Qvolutions-swaptions_ablation_QRCLin.ipynb** and **MerLin_IRIS_GreatViz-Qvolution2026.ipynb**

Besides this there are PennyLane + PyTorch models on different types of reservoirs tried on IRIS that have all been submitted.

Models studied:

Swaptions (MerLin — Regression), Classical Baselines:

- Persistence (predict tomorrow = today, naive baseline)
- Ridge Regression on PCA scores (classical linear)

- Classical MLP on PCA scores (classical nonlinear)
- Random Projection + Linear readout (sanity check)

Swaptions (MerLin — Regression), QRC Ablation Configs (testing various things inside the MerLin framework):

- Config D — QRC basic: 4 modes, 3 photons, 1 entangling layer, single reservoir
- Config E — QRC +photons: 4 modes, 4 photons, 1 entangling layer, single reservoir
- Config F — QRC +modes: 6 modes, 4 photons, 1 entangling layer, single reservoir
- Config G — QRC +entangle: 6 modes, 4 photons, 2 alternating entangling layers, single reservoir
- Config H — QRC ensemble: 6 modes, 4 photons, 2 alternating entangling layers, 5 independent reservoirs concatenated

For both datasets, two readout variants were tested on every QRC config:

- Linear readout (canonical QRC)
- MLP readout (diagnostic comparison)

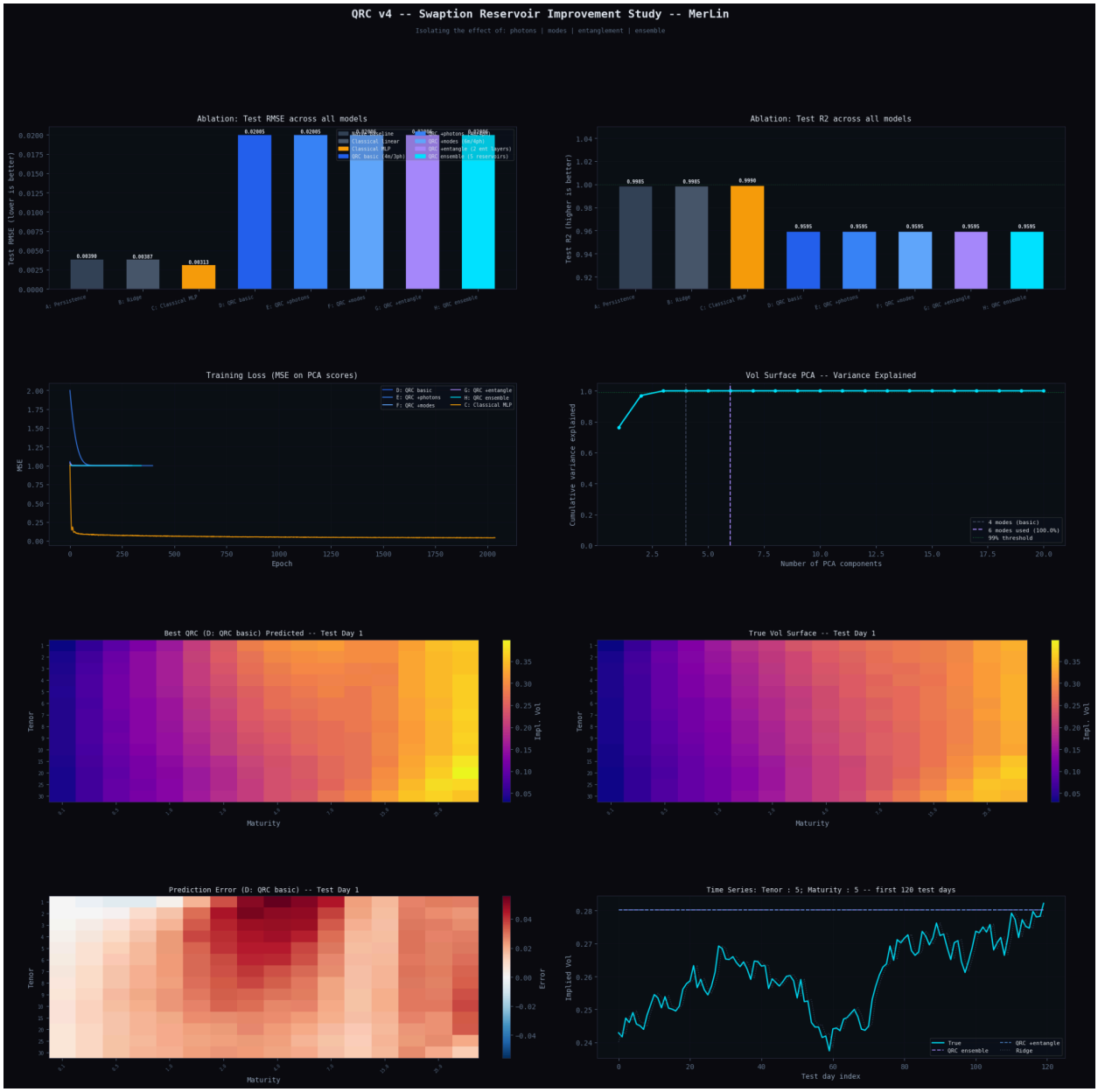
Plots and visualisations studied:

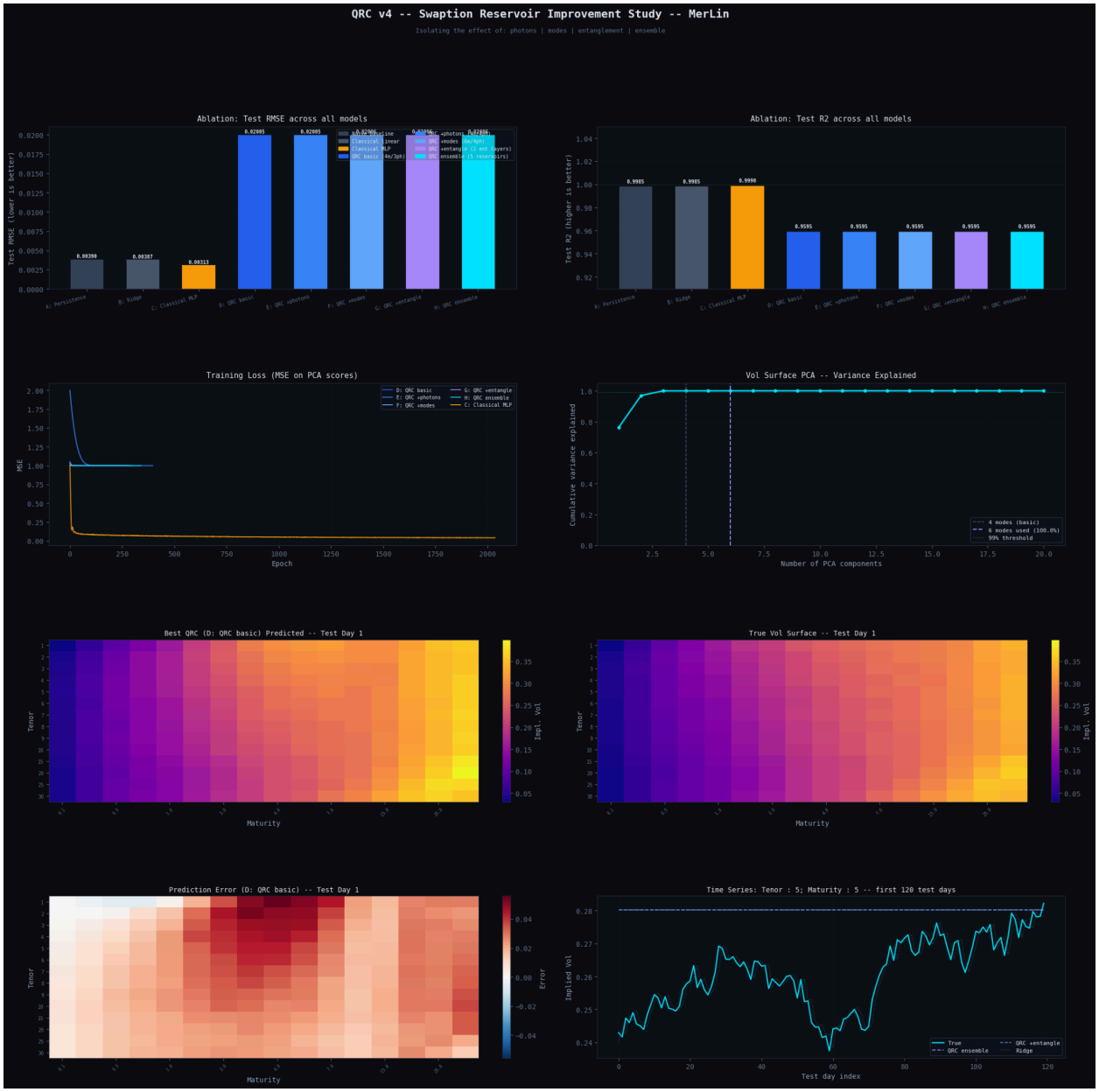
Swaptions — first ablation dashboard:

- RMSE bar chart across all models (train and test)
- R^2 bar chart across all models
- Training loss curves for trained models
- PCA cumulative variance explained curve
- Predicted vs true vol surface heatmap (test day 1)
- Prediction error heatmap (pred − true)
- Time-series comparison for a single vol cell (Tenor 5, Maturity 5) over first 100 test days

Swaptions — reservoir improvement dashboard:

- RMSE ablation bar chart: configs A through H
- R^2 ablation bar chart: configs A through H
- Training loss curves for all QRC variants + classical MLP
- PCA variance explained with 4-mode vs 6-mode comparison
- Predicted vs true vol surface heatmap (best QRC config, test day 1)
- Prediction error heatmap (best QRC config)
- Time-series comparison: QRC ensemble vs QRC +entangle vs Ridge vs true


QRC v4 -- Swaption Reservoir Improvement Study -- MerLin
Isolating the effect of: photons | modes | entanglement | ensemble
Ablation: Test RMSE across all models
Test RMSE (lower is better)
0.00390
0.00387
0.00313
0.02005
0.02005
A: Persistence
B: Ridge
C: Classical MLP
D: QRC basic
E: QRC +photons
F: QRC +modes
G: QRC +entangle
H: QRC ensemble
Ablation: Test R2 across all models
Test R2 (higher is better)
0.9985
0.9985
0.9990
0.9595
Training Loss (MSE on PCA scores)
MSE
Epoch
D: QRC basic
E: QRC +photons
F: QRC +modes
G: QRC +entangle
H: QRC ensemble
C: Classical MLP
Vol Surface PCA -- Variance Explained
Cumulative variance explained
Number of PCA components
4 modes (basic)
6 modes used (100.0%)
99% threshold
Best QRC (D: QRC basic) Predicted -- Test Day 1
True Vol Surface -- Test Day 1
Tenor
Maturity
Impl. Vol
Prediction Error (D: QRC basic) -- Test Day 1
Error
Time Series: Tenor : 5; Maturity : 5 -- first 120 test days
Implied Vol
Test day index
True
QRC ensemble
QRC +entangle
Ridge


QRC v4 -- Swaption Reservoir Improvement Study -- MerLin
Isolating the effect of: photons | modes | entanglement | ensemble
Ablation: Test RMSE across all models
Test RMSE (lower is better)
0.00390
0.00387
0.00311
0.02005
0.02005
0.02006
0.02006
0.02006
A: Persistence
B: Ridge
C: Classical MLP
D: QRC basic
E: QRC +photons
F: QRC +modes
G: QRC +entangle
H: QRC ensemble
Ablation: Test R2 across all models
Test R2 (higher is better)
0.9985
0.9985
0.9990
0.9595
0.9595
0.9595
0.9595
0.9595
Training Loss (MSE on PCA scores)
MSE
Epoch
D: QRC basic
E: QRC +photons
F: QRC +modes
G: QRC +entangle
H: QRC ensemble
C: Classical MLP
Vol Surface PCA -- Variance Explained
Cumulative variance explained
Number of PCA components
4 modes (basic)
6 modes used (100.0%)
99% threshold
Best QRC (D: QRC basic) Predicted -- Test Day 1
True Vol Surface -- Test Day 1
Tenor
Maturity
Impl. Vol
Prediction Error (D: QRC basic) -- Test Day 1
Error
Time Series: Tenor : 5; Maturity : 5 -- first 120 test days
Implied Vol
Test day index
True
QRC ensemble
QRC +entangle
Ridge